Exact certificates for quantum ordered search: seven queries for 850,000 elements and six for 91,000

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An exact quantum algorithm that searches an ordered list of n elements with k queries, applied recursively, searches N elements with $k\lceil \log_n N\rceil \le (k/\log_2 n)\log_2 N + k$ queries, so each pair (k,n) for which such an algorithm exists gives the coefficient $k/\log_2 n$. Carolan and Childs recently proved that exact algorithms with $(1/\pi)\ln n + o(\log n)$ queries exist for every sufficiently large n, so the optimal coefficient is $\ln 2/\pi \approx 0.2206$; their proof gives neither the threshold in n nor the lower-order term explicitly, and no finite pair (k,n). The finite pairs known so far come, beyond two queries, from numerical optimization, and to our knowledge none of those had been checked in exact arithmetic. We give exact integer feasibility certificates for eleven pairs: explicit polynomial chains that, by the characterization of Farhi, Goldstone, Gutmann and Sipser, prove that exact algorithms with these parameters exist. Each is checked by a standard-library verifier and by an independent checker built on FLINT. Seven queries search 850,000 elements, which gives at most $0.355383\log_2 N + 7$ queries for every N. To our knowledge this is the best explicit finite pair (k,n), that is, the smallest coefficient in a bound with explicit constants that holds for all N; the asymptotically optimal algorithms of Carolan and Childs have a smaller coefficient, but only for sufficiently large n with no explicit threshold. The coefficient improves on the numerical six-query algorithm for 90,000 elements of Wu *et al.* (0.364573) and on the five-query algorithm for 7265 elements (0.389811). Six queries search 91,000 elements, so with the infeasibility certificate of Wu *et al.* the largest list that six queries can search exactly satisfies $91000 \le N^*(6) < 94000$. We also certify Wu *et al.*'s six queries for 90,000 elements and four queries for 605 elements (the optimum found by Childs, Landahl and Parrilo) exactly. A certificate is a chain of symmetric Laurent polynomials in that characterization, with spectral factors that turn nonnegativity into a diagonal-dominance check. The chains were found by a smoothed max–min optimization over the values of the polynomials at roots of unity, where the chain conditions hold by construction. At $n = 850000$ the roots of the shifted polynomials come within 2.8×10^{-9} of the unit circle, and the spectral factors needed a local quadrature correction, a finer grid and multiplicative cepstral steps. The search, the certificates and this manuscript were produced by a language-model agent (Claude Opus 5.5, Anthropic).

I. INTRODUCTION

In the ordered search problem the input is a sorted list of n elements and a target, and the task is to find the position of the target. Equivalently, the input is a hidden string $x = 0^j 1^{n-j}$ and the algorithm must output j, using queries to the bits of x. Binary search uses $\lceil \log_2 n\rceil$ queries and is optimal classically. A quantum computer can do better by a constant factor, and the size of that factor has been studied for more than 25 years.

Lower bounds. After bounds by Buhrman and de Wolf [1], Farhi *et al.* [2] and Ambainis [3], Høyer, Neerbek and Shi proved that every exact quantum algorithm needs more than $(\ln n - 1)/\pi$ queries, about $0.2206\log_2 n$ [4]. Childs and Lee showed that the adversary method cannot give more, up to an additive constant [5].

Upper bounds. Farhi, Goldstone, Gutmann and Sipser (FGGS) introduced translation-invariant algorithms and found an exact three-query algorithm for $n = 52$, which gives $3\lceil \log_{52} N\rceil \approx 0.526\log_2 N$ queries [6]; for two queries the chain of their characterization is unique, and it is nonnegative exactly for $n \le 6$ [6, 7]. Høyer, Neerbek and Shi gave an algorithm of a different kind, a faster traversal of a binary search tree, with roughly $0.631\log_2 N$ queries [4]. Brookes, Jacokes and Landahl found three queries for $n = 56$ and four for $n = 550$, $4\log_{550} N \approx 0.439\log_2 N$, by conjugate-gradient descent [8] (unpublished; as cited in Ref. [7]), and Childs, Landahl and Parrilo (CLP) wrote the FGGS conditions as a semidefinite program and found the four-query optimum $n = 605$, $0.433\log_2 N$ [7]. Ben-Or and Hassidim gave a zero-error algorithm with about $0.32\log_2 N$ expected queries [9]. Carolan, Childs, Kovacs-Deak and Schaeffer (CCKS) proved that translation invariance loses nothing, characterized exact k-query algorithms by a polynomial program, and found the five-query optimum $n = 7265$ ($0.390\log_2 N$) with a linear-programming relaxation [10]. In 2026, Wu *et al.* reported "strong numerical evidence" from a matrix-free GPU semidefinite program that six queries search 90000 elements ($0.3646\log_2 N$), and certified that they cannot search 94000 [11].

Shortly afterwards Carolan and Childs settled the asymptotic question [12]. Their Theorem 4.14 gives, for every sufficiently large n, an analytic chain of the kind in Theorem 1 below with $k = (1/\pi)\ln n + o(\log n)$, so exact algorithms with that many queries exist and match the lower bound of Ref. [4] to leading order. Their Corollary 3.5 gives a zero-error algorithm with $(1/\pi)\ln n + O((\ln n)^{3/4})$ expected queries, whose coefficient is below that of Ben-Or and Hassidim. Both statements are asymptotic. For the exact algorithm neither the threshold in n nor the $o(\log n)$ term is explicit (the parameters come from existence statements, and the thresholds are only required to grow fast enough); for the zero-error algorithm the constants in the error terms are not given. No finite pair (k,n) is evaluated, and Ref. [12]

TABLE I. Exact ordered-search algorithms for finite (k, n): k queries search n elements, which gives at most $(k/\log_2 n)\log_2 N + k$ queries for N elements. "Published" lists the largest n reported before this work, with the method used; "certified here" lists the largest n with an exact feasibility certificate.

k	published n	$k/\log_2 n$	certified here	$k/\log_2 n$
3	56 [7, 8]	0.516587	56	0.516587
4	605, SDP [7]	0.432864	605	0.432864
5	7265, LP [10]	0.389811	5000	0.406911
6	90000, SDP [11][a]	0.364573	**91000**	**0.364220**
7	–	–	**850000**	**0.355383**

[a] Numerical evidence; $N^*(6) < 94000$ is certified [11]. We also certify $(6, 90000)$ exactly.

lists the extraction of algorithms for finite n as an open problem.

So the asymptotic coefficient is known, and what remains open is finite and explicit: the largest $n = N^*(k)$ that k queries can search exactly. Every certified pair (k, n) proves that N elements can be searched exactly with $k\lceil\log_n N\rceil \le (k/\log_2 n)\log_2 N + k$ queries. We compare pairs by the coefficient $k/\log_2 n$, which we round up in the sixth decimal so that every quoted bound holds for all N; every bound $c\log_2 N$ quoted for a pair with k queries carries the additive k. The largest values known before this work were $n = 6$, 56, 605 and 7265 for $k = 2, \ldots, 5$, and $90000 \lesssim N^*(6) < 94000$. The two-query value follows from an exact computation [6, 7]; those for $k \ge 3$ were found numerically, and to our knowledge none of the algorithms behind them had been checked in exact arithmetic.

This paper gives exact integer feasibility certificates for eleven pairs (k, n) (Tables I and III). The main results are:

- **Seven queries search 850,000 elements**, so ordered search on N elements needs at most $7\lceil\log_{850000} N\rceil \le 0.355383\log_2 N + 7$ queries. To our knowledge this is the best explicit finite pair (k, n) (Sec. VI compares it with Ref. [12]). Its coefficient is 2.5% below that of the numerical $(6, 90000)$ and 8.8% below that of $(5, 7265)$; a seven-query algorithm needs $n \ge 602490$ to beat $(6, 90000)$. We also certify $(7, 650000)$ (0.362505), the first certified seven-query algorithm, $(7, 700000)$ (0.360509) and $(7, 775000)$ (0.357804).
- **Six queries search 91,000 elements**, which goes beyond the numerical record. With the certified upper limit of Ref. [11], which we did not re-check, $91000 \le N^*(6) < 94000$.
- **Exact certificates of earlier numerical algorithms**: $(6, 90000)$ of Wu *et al.*, which turns their numerical evidence into a theorem, and $(4, 605)$, the optimum of CLP; also $(3, 56)$, $(5, 5000)$, $(6, 45000)$ and $(6, 80000)$.

The rest of the paper is organized as follows. Section II recalls the polynomial characterization and restates it in value form. Section III describes the certificate and the two checkers, Sec. IV how the chains were found, and Sec. V how a floating-point chain becomes an exact certificate. Section VI gives the results, Sec. VII the limits of the method and open problems, and Sec. VIII how the work was done.

II. THE POLYNOMIAL CHARACTERIZATION

A Laurent polynomial $q(z) = \sum_{|j|<n} c_j z^j$ is symmetric if $c_{-j} = c_j \in \mathbb{R}$; on the unit circle it is the cosine polynomial $q(\theta) = c_0 + 2\sum_{j=1}^{n-1} c_j \cos j\theta$. The Fejér kernel is $F_n(z) = \sum_{|j|<n}(1 - |j|/n)z^j$, so $F_n(\theta) = \sin^2(n\theta/2)/(n\sin^2(\theta/2))$.

Theorem 1 (FGGS [6]; CLP [7]; Theorem 8 of [10])**.** *An exact k-query translation-invariant quantum algorithm for n-element ordered search exists if and only if there are symmetric Laurent polynomials $q_0, \ldots, q_k$ of degree less than n, nonnegative on $|z| = 1$, such that*

$$q_0 = F_n, \qquad q_k = 1, \qquad c_0(q_t) = 1 \quad (t = 0, \ldots, k), \tag{1}$$

$$q_t(z) = q_{t-1}(z) \quad \text{if } z^n = (-1)^t \qquad (t = 1, \ldots, k). \tag{2}$$

By Theorem 1 of Ref. [10], translation-invariant algorithms are optimal, so $N^*(k)$ is the largest n for which such a chain exists, and feasibility is monotone in n [10]. We do not construct the unitaries of the algorithms; the theorem guarantees that they exist. The deliverables of this paper are therefore exact feasibility certificates: explicit chains that prove that exact algorithms with the stated (k, n) exist. This is what we mean by a certified algorithm.

Value form. Let $\theta_m = \pi m/n$, $m = 0, \ldots, n$. The points $e^{\pm i\theta_m}$ with m even are the n roots of $z^n = 1$, and those with m odd are the n roots of $z^n = -1$. A symmetric Laurent polynomial of degree less than n is a cosine polynomial with $c_n = 0$; it is determined by its $n + 1$ values $v_m = q(\theta_m)$, and by the type-1 discrete cosine transform

$$c_0 = \frac{1}{n}\sum_{m=0}^{n} w_m v_m, \qquad c_n = \frac{1}{n}\sum_{m=0}^{n} w_m (-1)^m v_m = 0, \tag{3}$$

with $w_0 = w_n = \frac{1}{2}$ and $w_m = 1$ otherwise. So $c_0 = 1$ and $c_n = 0$ hold exactly when the values on each parity class average to one over the corresponding n roots of unity. Conditions (2) say that q_t and q_{t-1} share their values on one parity class, alternating with t. A chain is therefore a path of "half-vectors"

$$u_0 \to u_1 \to \cdots \to u_{k-1}, \tag{4}$$

where u_t lives on the parity class of $t+1$ (odd m for u_0), u_0 is the odd-m part of F_n, $u_0(m) = 1/(n\sin^2(\pi m/2n))$, $u_{k-1} = 1$, every u_t has mean one, and q_t interleaves u_{t-1} and u_t (q_0 itself is n at $m = 0$ and 0 at the other even m). Conditions (1) and (2) then hold by construction, and the problem is to choose $u_1, \ldots, u_{k-2}$ so that every interleaved q_t is nonnegative on the whole circle, not only at the samples. All of the difficulty lies between the samples: the interleaving makes q_t oscillate at frequency close to n, with narrow dips between neighboring samples.

III. CERTIFICATES AND CHECKERS

A certificate lists, for $t = 1, \ldots, k-1$, the integer coefficients of q_t over a common denominator D divisible by n, and the integer coefficients of a polynomial $h_t(z) = \sum_{j=0}^{n-1} h_{t,j} z^j$ over a denominator H. The verifier fixes $q_0 = F_n$ and $q_k = 1$ and checks in exact integer arithmetic:

1. every q_t has constant term 1;

2. *root conditions.* On $z^n = s$ we have $z^{-j} = s\, z^{n-j}$, and $1, z, \ldots, z^{n-1}$ are linearly independent as functions on those n roots. So with $d = q_t - q_{t-1}$, condition (2) holds if and only if $d_j + s\, d_{n-j} = 0$ for $j = 1, \ldots, n-1$, where $s = (-1)^t$;

3. *nonnegativity.* The residual $r_t = q_t - |h_t|^2$, a symmetric Laurent polynomial, must be diagonally dominant: $r_{t,0} \geq 2\sum_{j\geq 1} |r_{t,j}|$.

The last condition implies $q_t(\theta) \geq |h_t(e^{i\theta})|^2 + r_{t,0} - 2\sum_{j\geq 1} |r_{t,j}| \geq 0$ for all θ. We call $(r_{t,0} - 2\sum_{j\geq 1} |r_{t,j}|)$ the certified margin of q_t. A positive margin is always achievable: by the Fejér–Riesz theorem [13, 14] a strictly positive $q_t - \gamma$ equals $|h|^2$ for a polynomial h of degree $n-1$, and rounding h perturbs $|h|^2$ by much less than γ. The autocorrelation $|h_t|^2$ is computed exactly by Kronecker substitution, that is, by one multiplication of two large integers. The verifier (`verifier/verify.py`, Python standard library only) is pinned by a hash, version 1.1. Besides the verdict it raises an alarm if a certificate contradicts a proven or published bound: $k < \ln(n)/\pi - 2$ (the bound of Ref. [4], weakened), or n above the published feasibility limits 6, 605, 7265 and 93999 for $k = 2, 4, 5, 6$. Either alarm would point to a bug in the verifier or an error in the literature. Its test suite contains the certificates for $(3, 56)$ and $(4, 605)$, every verdict path, 17 malformed or tampered inputs, and cross-checks of the Kronecker autocorrelation and of the root conditions against brute force.

An independent checker shares no code with the verifier and uses a different route for each condition, in rational arithmetic from FLINT [15] via python-flint. For the root conditions it checks that $z^n - s$ divides the integer polynomial $z^{n-1}(q_t - q_{t-1})(z)$; it forms $|h_t|^2$ as the product $h_t(z) \cdot z^{n-1} h_t(1/z)$ and repeats the diagonal-dominance test. Every acceptance condition raises an explicit error, so the checks do not depend on Python assertions, which `python -O` removes; entries must be exact integers and both denominators positive. Its test suite feeds it 26 malformed or tampered inputs and repeats the command-line rejections under `python -O`. Both programs check certificates; they do not limit the size of the input and are not meant as services for hostile input. On all eleven certificates the two checkers agree to the last printed digit on every margin. An audit script records both results with the verifier's hash, and adds a floating-point evaluation of every q_t on a grid at least 32 times finer than n (the next power of two above $32n$) as a sanity check. One difference in presentation: the frozen verifier prints $k/\log_2 n$ rounded to nearest (0.355382 for $(7, 850000)$, 0.357803 for $(7, 775000)$), whereas the coefficients quoted in this paper are rounded up.

IV. FINDING THE CHAINS

Reduced basis. Each free half-vector u_t ($t = 1, \ldots, k-2$) is parametrized by its individual samples for $m \leq m_0 = 100$ and, beyond, by piecewise-linear hat functions in $\log\theta$ with knots spaced by a ratio of at most $\rho = 1.05$ up to $\theta = \pi$. This gives 708 variables at $(6, 45000)$, 766 at $(6, 90000)$ and $(6, 91000)$, and 1158 at $(7, 650000)$, 1168 at $(7, 700000)$, 1178 at $(7, 775000)$ and 1188 at $(7, 850000)$.

Weighted max–min. The natural scale of the chain is the Fejér tail, so we maximize the weighted margin $\min_{t,\theta} q_t(\theta)/\omega(\theta)$ with $\omega(\theta) = \min\{1, 1/(n\sin^2(\theta/2))\}$. The minimum is replaced by the smooth $\mathrm{softmin}_\mu(x) = -\mu \log \sum_i e^{-x_i/\mu}$, which lies within $\mu \ln(\#\text{points})$ of the true minimum, and is taken over two sets of points: a grid 8 times finer than n (evaluated by batched type-1 discrete cosine transforms), and "extra points" at the local minima of q_t found in earlier passes, which catch the narrow dips that the grid misses. Samples where u_t is fixed (those of u_0 and $u_{k-1} = 1$) are masked: on the samples of u_0 the ratio is exactly $q/\omega = 1$, which would cap the objective. We maximize with L-BFGS-B [16, 17] and continuation in μ from 1 to 0.003.

Passes. After each optimization, every local minimum of every q_t on a 16-fold grid is refined by five Newton steps, with q_t' and q_t'' interpolated from their own oversampled grids. The minima that fall below the current objective become extra points, and the pass is repeated until none is added. The smallest refined value of q_t/ω over all t is the "true minimum" of the chain; a chain is handed to the certificate builder when it is positive. An earlier version refined only the minima whose parabolic estimate from the grid looked low. At large θ, where $\omega \sim 1/n$ is smaller than the error of that estimate, it skipped a band of 28 real negative minima in the $(6, 90000)$ chain. The certificate builder detected them, because the chain was negative on its finer grid, and after the fix every grid minimum is refined.

Large n. For $k = 7$ three changes keep a pass near $O(n \log n)$: extra points are sparse rows that interpolate the oversampled grid with a Gaussian-windowed sinc, in the spirit of nonuniform FFTs [18], instead of dense rows of length n; local minima are refined from interpolated derivative grids; and the $k - 1$ transforms of one evaluation run as one threaded batch, 0.6 s per objective and gradient at $n = 650000$. The decisive change was a log parametrization: every block is written as $a = n\,e^x/(r \cdot e^x)$, where $r \cdot a = n$ is the mean-one constraint. It keeps $a > 0$, makes the constraint exact and turns optimizer steps into relative changes, which matters because the samples of a chain span many decades. With a linear diagonal scaling instead, the weighted grid minimum after 1000 iterations at $\mu = 1$ was -800; with the log parametrization it was -53.

Warm starts. At fixed sample index m (not fixed θ) the shape of a chain changes slowly with n. When the basis has the same number of variables at two sizes, we copy the coefficients, rescale each block to mean one, and map extra points by $\theta \mapsto \theta\, n_{\text{old}}/n_{\text{new}}$. The $(6, 90000)$ chain moved to $n = 91000$ this way starts with true minimum -3.4, against -531 after a cold first pass. When the number of variables changes, as from $(7, 650000)$ (1158) to $(7, 700000)$ (1168), each coefficient beyond m_0 is stored as a ratio to the envelope $U^{1-t/(k-1)}$, where $U(\theta) \approx 1/(n \sin^2(\theta/2))$ is the envelope of u_0, and carried over at the same fraction of the knot range (fixed m at the bottom of the basis, fixed θ at $\theta = \pi$). The $(7, 700000)$ start then has true minimum -129, against -874 when the samples are copied at fixed m; the $(7, 775000)$ start, from the 700000 chain, has -181, and the $(7, 850000)$ start, from the 775000 chain, -266.

Table II lists the runs behind the certificates; all ran on a laptop (Apple M4, 16 GB of memory).

At $(7, 650000)$ the weighted minimum over the grid and the extra points crossed zero in the 7th pass. The true minimum became positive in the 18th pass and reached 0.987 in the 22nd and last, by which time 1.65 million exact local minima served as constraints. Warm-started from that chain, the run at $(7, 700000)$ had a positive minimum over the grid and the extra points already at the first continuation stage $\mu = 0.1$ (cold at 650000: -5.0), and reached a true minimum of 0.990 after 6 passes and 643,058 extra points. From that chain the run at $(7, 775000)$ needed 20 passes (12.4 hours, 776,661 extra points) to reach 0.985, and from there the run at $(7, 850000)$, where a cold start had stalled (Sec. VII), needed 22 passes (19.8 hours, 946,854 extra points) to reach 0.977.

TABLE II. Search runs. "Passes" counts optimization passes, and "true min" is the smallest refined local minimum of q_t/ω over all t at the end of the run. For $(6, 90000)$, $(6, 91000)$, $(6, 92000)$, $(7, 700000)$, $(7, 775000)$ and the warm-started $(7, 850000)$ it comes from the corrected minimum check of Sec. IV, which refines every grid minimum. For $(6, 45000)$, $(6, 80000)$ and $(7, 650000)$ it is the value reported by the original run, whose check refined only the minima with a low parabolic estimate; these three were not recomputed. The values are diagnostics of the search: positivity of every certified chain is proved by its certificate. The $(6, 45000)$ and $(6, 80000)$ chains were found by an earlier dense implementation of the same search. Two unsuccessful runs are listed at the bottom.

k	n	variables	passes	time	true min
6	45000	708	6	13 min	0.994
6	80000	754	12	2.2 h	0.808
6	90000	766	40 + 40	1.8 h + 84 min[a]	0.265
6	91000	766	14	35 min[b]	0.147
7	650000	1158	22	16.1 h	0.987
7	700000	1168	6	3.8 h[b]	0.990
7	775000	1178	20	12.4 h[b]	0.985
7	850000	1188	22	19.8 h[b]	0.977
6	92000	766	40	1.9 h	-0.17
7	850000	1188	4	7 h	≈ -25[c]

[a] The run was resumed after the minimum check was fixed (Sec. IV).
[b] Warm start from the chain in the row above.
[c] Cold start; weighted grid minimum, the same on every q_t; a basis twice as fine (2200 variables) reached -23.8 in two further passes.

V. FROM A FLOATING-POINT CHAIN TO AN EXACT CERTIFICATE

Exact rounding. The coefficients are rounded to the denominator $D = n \cdot 2^{57}$ step by step. Each increment $q_t - q_{t-1}$ is projected in integers onto the subspace allowed by (2), so the root conditions hold exactly. The last polynomial q_{k-1} is not rounded but solved for: its values on both parity classes are fixed, by q_{k-2} and by $q_k = 1$.

Spectral factors. For each q_t we take γ_t to be half the minimum of q_t on a fine grid and compute the minimum-phase spectral factor h_t of $q_t - \gamma_t$, rounded to the denominator $H = 2^{60}$. We start from the cepstrum [19] on a half-spectrum FFT grid of M points, apply truncated Newton steps of Wilson's method [20], and double M up to 2^{26} until $1/|h_t|^2$ is resolved. If the factor error is still above the budget left by γ_t, mixed-precision Newton refinement follows, with the residual $q_t - |h_t|^2$ computed exactly by FLINT.

Spikes. At $n = 650000$ the roots of $q_t - \gamma_t$ come within 1.4×10^{-8} of the unit circle, closer than the grid spacing $2\pi/2^{26} = 9.4 \times 10^{-8}$. Then $1/|h_t|^2$ has spikes narrower than the grid, and plain Newton refinement oscillates: the first attempt failed with a certified margin of -7.7×10^{-7} on q_3, and both checkers rejected it. The fix computes the Fourier coefficients of the Newton step exactly near each spike. Around every minimum where the estimated root distance $\sqrt{2(q_t - \gamma_t)/q_t''}$ is below 3.5 grid spacings, a smooth partition of unity isolates the spike. The spike's integral is computed by a tan-mapped

Gauss rule on the core and Gauss panels on the transition, minus the grid's own share. The corrections are then folded into all n coefficients by a 26-term moment expansion and the transpose of the windowed-sinc interpolation. On a test chain this cut the coefficient error from 0.22 to 3×10^{-10}. With it, q_5 (18,304 spikes) converged in two steps and q_6 (56,639 spikes, plain-FFT error 3.7×10^{-6}) in eight. The certificate took 22 minutes and 6.6 GB to build. The $(6, 91000)$ certificate needed the same correction for 32,504 spikes in q_5, with roots 1.2×10^{-8} from the circle, and took 3 minutes.

A finer starting grid. At $(7, 700000)$ the correction was not enough for q_6: its 70,223 spikes come within 8.7×10^{-9} of the circle, the cepstrum on 2^{26} points left a factor error of 1.2×10^{-5}, and the first Newton step overshot at the spikes. From there the residual fell only by a factor of 4 per step, as for Newton's method on $x^2 = a$ started at $x \gg \sqrt{a}$, and after 10 steps it was 1.7×10^{-6}. Both checkers rejected the certificate (margin -0.33 on q_6). Recomputing that one factor on 2^{27} points gave a starting error of 2.6×10^{-11}, and two refinement steps reached 7.5×10^{-16} (6 minutes, with the other factors kept). The builder now retries on a grid twice as fine whenever the refinement does not converge. At $(7, 775000)$ three of the six factors needed this retry: on 2^{26} points the refinement of q_1, q_5 and q_6 ended with the residual roughly doubling at every step, and on 2^{27} points all three converged, q_6 (90,040 spikes, closest root 5.2×10^{-9} from the circle) in nine steps. The refinement now stops once its residual has grown twice in a row, keeps the best factor, and may take up to 20 steps. Most of the 6.7 hours (peak memory 11 GB) went into the spike corrections: in q_2 nearly every minimum of the oscillation is a spike (386,979 of them), and each of its six refinement steps took about 30 minutes.

Cepstral steps. At $(7, 850000)$ the retry was not enough for q_6. Its 130,228 spikes come within 2.8×10^{-9} of the circle, the cepstrum on 2^{27} points left a factor error of 8.7×10^{-6}, and the refinement stalled at a residual of 1.3×10^{-7}; a grid of 2^{28} points would need about 25 GB, more than the laptop has. After a build of 10.5 hours (peak memory 12.4 GB) both checkers rejected the certificate (margin -0.031 on q_6). The quadrature was not the cause: on this factor the tan-mapped rule agrees with a much finer graded rule to 1.4×10^{-10} of the largest coefficient. Rather, at about 300 spikes the stalled iterate had no near-zero at all, with $|h_t|^2$ up to 3000 times above $q_t - \gamma_t$. Wilson's step is the linearization of

$$h \leftarrow h \exp\Big\{ \log \frac{q_t - \gamma_t}{|h|^2} \Big\}_+ , \tag{5}$$

where $\{\cdot\}_+$ keeps half of the constant term and the positive powers of z, and as a linear step it cannot bring a root close to the circle in one step. Equation (5) itself gives the exact minimum-phase factor h_t in one step whenever h is minimum phase, because $\log(h_t/h)$ is then analytic in the unit disk. Since the product has degree below n, only the first n coefficients of the logarithm are needed, and the exponential is taken as a power series modulo z^n. Those coefficients come from the grid plus the spike correction above, here with Gauss panels graded geometrically from half the root distance out to the window, since $\log(q_t - \gamma_t)$ has dips of every width in between. From $h = 1$, which is minimum phase, two steps on 2^{27} points gave residuals of 1.6×10^{-11} and 5.7×10^{-17} (27 minutes, 9.2 GB, with the other factors kept). From the stalled iterate the same step failed, because that iterate has near-zeros where $q_t - \gamma_t$ has none. The builder now falls back on these steps after the retry, and a rerun reproduces this h_6 bit for bit.

TABLE III. The eleven certificates; each gives at most $(k/\log_2 n)\log_2 N + k$ queries for N elements. The margin is the smallest certified margin over t (absolute, in units of the coefficients of q_t); both checkers report the same value. Times are for the standard-library verifier and the FLINT checker on the laptop of Table II (as recorded in the audit file; for $(7, 700000)$ and $(7, 775000)$ both checkers ran alongside another search); files are gzipped JSON except the two smallest.

k	n	$k/\log_2 n$	margin	size	verifier	FLINT
3	56	0.516587	2.4×10^{-3}	4.8 kB	< 0.1 s	< 0.1 s
4	605	0.432864	4.3×10^{-5}	74 kB	< 0.1 s	< 0.1 s
5	5000	0.406911	1.0×10^{-4}	0.35 MB	0.2 s	< 0.1 s
6	45000	0.388158	1.1×10^{-5}	3.9 MB	5.5 s	0.5 s
6	80000	0.368376	6.1×10^{-6}	6.8 MB	13 s	1.0 s
6	90000	0.364573	1.7×10^{-6}	7.6 MB	18 s	1.4 s
6	91000	0.364220	9.3×10^{-7}	7.7 MB	21 s	1.6 s
7	650000	0.362505	7.5×10^{-7}	64.4 MB	7.1 min	12 s
7	700000	0.360509	7.1×10^{-7}	68.8 MB	10 min	19 s
7	775000	0.357804	6.5×10^{-7}	75.5 MB	12 min	23 s
7	850000	0.355383	5.9×10^{-7}	82.5 MB	12 min	16 s

VI. RESULTS

Table III lists the certificates and Fig. 1 compares the coefficients with the published ones and with the asymptotic optimum.

Seven queries. The certificate for $(7, 850000)$ proves that ordered search on N elements can be solved exactly with $7\lceil \log_{850000} N \rceil \le 0.355383 \log_2 N + 7$ queries. To our knowledge this is the best explicit finite pair (k, n): the smallest coefficient in a bound with explicit constants that holds for all N. Its coefficient is still a factor 1.61 above the optimum $\ln 2/\pi$ [4, 12], and it is not the best exact algorithm in the asymptotic sense. By Theorem 4.14 of Ref. [12], exact algorithms whose coefficient tends to $\ln 2/\pi$ exist for all sufficiently large n, and a proof is as good as a certificate. But neither the threshold nor the lower-order term is explicit there, so the statements of Ref. [12] determine neither a finite pair with $k/\log_2 n < 0.355383$ nor the N beyond which that

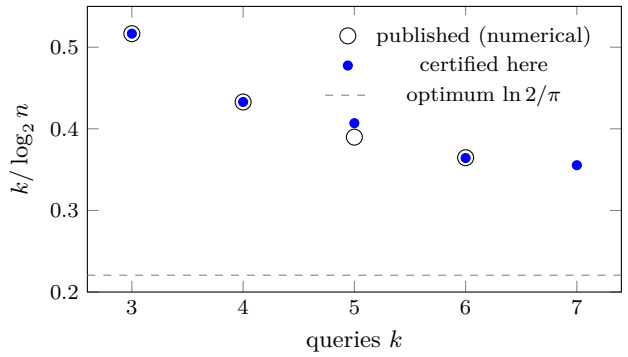


FIG. 1. Coefficients $k/\log_2 n$ of exact algorithms for finite (k, n), each of which gives at most $(k/\log_2 n)\log_2 N + k$ queries: the largest published n for each k [7, 10, 11] and the largest certified here. The dashed line is the asymptotic optimum $\ln 2/\pi = 0.2206$ [4, 12].

construction uses fewer than $0.355383\log_2 N + 7$ queries. The one explicit count in Ref. [12] belongs to its zero-error algorithm, which is not exact: each attempt uses $Q = \lceil (\ln n + 2(\ln n)^{3/4})/\pi \rceil$ queries plus a constant, and the expected number is that divided by a success probability $1 - O((\ln n)^{-1/2})$ with an unspecified constant. Even Q alone is below $0.355383\log_2 n$ only for $\ln n > 115$, that is, $n \gtrsim 10^{50}$. The certificates for $(7, 650000)$, found first, $(7, 700000)$ and $(7, 775000)$ give $0.362505\log_2 N + 7$, $0.360509\log_2 N + 7$ and $0.357804\log_2 N + 7$.

Six queries. The certificate for $(6, 91000)$ goes beyond the numerical record 90000 of Ref. [11]. Together with their certified infeasibility at 94000, which we did not recheck, and with the optimality of translation-invariant algorithms and monotonicity [10], $91000 \le N^*(6) < 94000$. The certificate for $(6, 90000)$ makes their numerical result rigorous. The sequence of six-query certificates 45000, 80000, 90000, 91000 shows the margin shrinking as n approaches the limit of the search (Table III).

Smaller k. The $(4, 605)$ certificate confirms exactly that four queries suffice for 605 elements, the optimum of CLP [7]. The cutting-plane linear program we used for calibration reproduces the known thresholds numerically: two queries 6 but not 7, three queries 56 but not 57, four queries 605 but not 606 (optimal weighted margin $+8.7\times 10^{-5}$ at 605 and -4.2×10^{-4} at 606). The reduced basis is nearly lossless at these sizes: it is feasible at exactly 605 and infeasible at 606, and at $k = 5$ its best relative margin at the true limit 7265 is -0.004. We did not certify $(5, 7265)$; the $(5, 5000)$ certificate is a calibration.

VII. LIMITS AND OPEN PROBLEMS

How far the search reaches. At $k = 6$ the same search ended at a true minimum of -0.17 for $n = 92000$ after 40 passes, with the minimum over grid and extra points still rising slowly. Its limit is therefore near 91500–92000, within about 3% of the upper limit 94000. The true minima of the certified six-query chains show how the margin behaves near the limit (Table II). Far from the limit it stays close to its ceiling, $q_1/\omega = 1$ at the fixed samples of u_0 (0.994 at $n = 45000$); nearer the limit it falls steeply (0.808 at 80000, 0.265 at 90000, 0.147 at 91000). The seven-query chains still have 0.987 at 650000, 0.990 at 700000, 0.985 at 775000 and 0.977 at 850000, so even 850000 is likely far from $N^*(7)$. At $k = 7$ the gap is larger. The ratios $N^*(k)/N^*(k-1)$ are 9.3, 10.8, 12.0 and between 12.5 and 12.9 for $k = 3, \ldots, 6$. The bounds of Refs. [4, 12] give $\ln N^*(k) = \pi k + o(k)$, so the geometric mean of these ratios tends to $e^\pi \approx 23.1$, but successive ratios need not converge: $\ln N(k) = \pi k + c(-1)^k$ with $0 < c < \pi/2$ is increasing, has the same asymptotics, and its ratios alternate. As a heuristic only, continuing the slow growth of the observed ratios suggests $N^*(7) \approx 1.1$–1.2 million. A cold run at $n = 850000$ stalled at a weighted grid minimum of about -25, the same on every q_t and at every frequency. Refitting that chain in a basis twice as fine ($m_0 = 200$, $\rho = 1.025$, 2200 variables) moved the minimum only to -23.8 in two passes. So the resolution of the basis is not what limits the search; the optimizer is. In the coefficients the restricted problem is a concave max–min, since each q_t is linear in them, and the softmin is within $\mu\ln(\#\text{points}) \approx 17.5\mu$ of its optimum. Yet in the cold first pass at $n = 850000$ every stage of the continuation ended far below that bound (at $\mu = 1$, -106, although -22 is attainable). A solver that converges faster on the restricted linear program, for instance stabilized cutting planes with an interior-point method, is one route to larger n. The other is a ladder of warm starts, which took the present solver from 650000 to 700000 in 3.8 hours, on to 775000 in 12.4 hours and to 850000 in 19.8 hours, with the margin still near its ceiling. In the same basis in which the cold run stalled, the warm chain at 850000 is feasible, which confirms that the optimizer was the limit. At that size the bottleneck was the certificate builder, not the search (Sec. V): 2^{27} points is the finest grid that fits in memory.

Size of the certificates. A certificate stores $2(k-1)n$ large integers, so the $(7, 850000)$ file has 82.5 MB compressed and takes 12 minutes to check with the standard library. Larger k will need either a more compact format, for instance coefficients generated from a short description, or a certificate that exploits the structure of the chain.

Open problems. (i) Close the gap $91000 \le N^*(6) < 94000$, and certify the five-query optimum $(5, 7265)$, which sits at the limit of feasibility, so its margins are tiny. (ii) Certify infeasibility exactly, for example at $(5, 7266)$ or $(6, 94000)$, with a rational dual certificate. (iii) Explicit finite instances of the Carolan–Childs construction [12], to compare its finite-n performance with $N^*(k)$; Ref. [12] lists this as open too. (iv) An explicit finite pair with $k/\log_2 n < 0.32$. That is the expected-query coefficient of the zero-error algorithm of Ben-Or and Hassidim [9], and it is below the coefficient of every explicit finite pair known, including ours. Asymptoti-

cally the question is closed: by Ref. [12] exact algorithms reach $0.2206\log_2 n$ to leading order (Theorem 4.14), and its zero-error algorithm (Corollary 3.5) has a smaller coefficient than that of Ref. [9]. Only the explicit finite version is open. Seven queries would need $n > 2^{7/0.32} \approx 3.8$ million, which is above our heuristic estimate of $N^*(7)$, so such a pair probably needs more than seven queries.

VIII. HOW THE RESULTS WERE FOUND

The problem was set up on 28–29 October 2026 by a language-model session (Claude Opus 5.5, Anthropic), after a human operator asked it to start a new problem from a list of verifiable open problems and solve it. The session wrote the verifier, its tests and the record snapshot, and then carried out the search alone. The literature check that found Refs. [11, 12] came after the pipeline was built. The verifier was then revised once, to version 1.1: the record it compares against changed from $(5, 7265)$ to the numerical $(6, 90000)$, and the alarm limit 93999 for $k = 6$ was added. All certificates were audited under version 1.1 with matching hashes. The independent checker was written by the same session, but with a different route for each condition and a different arithmetic library. The certificates for $k \le 6$ up to $n = 80000$ were audited on 29 October, $(7, 650000)$ on 30 October after a 16-hour search, $(6, 90000)$, $(6, 91000)$ and $(7, 700000)$ on 31 October, $(7, 775000)$ on 1 November after a 12-hour search warm-started from the 700000 chain, and $(7, 850000)$ on 2 November after a 20-hour search warm-started from the 775000 chain, once its first certificate, rejected by both checkers, had been repaired (Sec. V). The human operator chose the problem, asked for the seven-query run and for larger n before a paper was drafted, decided when to commit results and approved the publication of the certificates. This manuscript was drafted by the same model from the results summary, the audit records and the run logs, and checked against them. After an external review, the ratio extrapolation of Sec. VII was marked as a heuristic, the finite-query bounds were stated with their additive k, the results were described consistently as feasibility certificates, and the independent checker was hardened (Sec. III). A second review re-checked all eleven certificates with a third, independently written checker and found the same margins. After it the comparison with Ref. [12] was corrected (the headline claim is the best explicit finite pair, not the best exact algorithm, and open problem (iv) was restated), and several counts and times were brought into line with the logs and the audit file; that revision of the text was made by a later language-model session (Claude Fable 5.2, Anthropic) and involved no new computation.

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DATA AVAILABILITY

The eleven certificates, the frozen verifier with its tests and hash pin, the independent checker with its tests, the audit records and a manifest with the SHA-256 sum of every released checker, test, program, log and data file (`MANIFEST.sha256`) are available at https://github.com/maxxfan/aixmathxphysics/releases/tag/ordered-search-v5. The four seven-query certificates (64.4, 68.8, 75.5 and 82.5 MB) are attached to that release as assets. Running `python3 verifier/verify.py` on a certificate re-checks it with the Python standard library, and `results/check_independent.py` (python-flint) rechecks it independently. The search and certificate-building programs (`work/lead/`, `qlib/`) and the run logs from which the pass counts, times, spike counts and residuals of Secs. IV and V are taken (`results/logs/`) are part of the same release; they are not needed to check a certificate.

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